Weil Fingerprint Orthogonality and the Flat Schmidt Spectrum of Conjugate Pairs

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June 2026

Abstract

The companion entanglement paper established numerically that the normalized Gram–Schmidt admission residual satisfies $w_j = 1$ for every admitted Weil fingerprint direction, across $q \in \{29, 61, 101, 151\}$ and multiple generic blocks. The equality was left as an open analytic strengthening. We close it here by explicit computation. The k=3 Weil fingerprint of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for any block (c_1, c_2, c_3) is, up to a global phase, the discrete Fourier mode $\hat{u}_A = (e^{2\pi i A t/q})_{t=0}^{q-1}$ at frequency $A = c_1 b_1 + c_2 b_2 + c_3 b_3 \bmod q$, where (b_1, b_2, b_3) are the b-components of the three path elements. Since distinct Fourier modes are orthogonal, any newly admitted fingerprint is already orthogonal to the current span, giving $w_j = \|w_j^{\mathrm{GS}}\|/\|v_j\| = 1$ exactly. For matched single-character blocks, the conjugate pair produces sign-reflected frequency sets, so the paired Fourier supports are canonically identified and the purity hypothesis of the companion paper is derived. For general canonical blocks, the equal-count identity is recovered, while the literal Schmidt pairing remains conditional on a canonical block identification. Combined, these results prove analytically that the Schmidt spectrum of the closed conjugate pair is flat on the residual support, and the reduced entropy is $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$, where $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ is the residual rank.

Keywords: Weil representation; Heisenberg group; k-step fingerprints; Fourier modes; Gram–Schmidt orthogonality; conjugate pairs; Schmidt spectrum; entanglement entropy.

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1 Introduction

The spectral admissibility cascade of the Cosmochrony programme operates through Gram–Schmidt projections of Weil fingerprint vectors on the Cayley graph $\mathrm{Cay}(\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$ [3, 1]. The companion entanglement paper [**?**] established that the reduced entropy of a closed conjugate Weil pair equals the logarithm of the residual fibre-level rank,

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n),$$

under the hypothesis that the normalized Gram–Schmidt admission residual satisfies $w_j = 1$ for every admitted direction. This equality was verified numerically across $q \in \{29, 61, 101, 151\}$ and multiple generic blocks, and was left as an open analytic strengthening.

The present note closes it. The proof is a direct computation: the k=3 Weil fingerprint, as implemented in the spectral pipeline, is always proportional to a discrete Fourier mode on $\mathbb{Z}/q\mathbb{Z}$. Since distinct Fourier modes are orthogonal, any newly admitted fingerprint is already orthogonal to the current span, making the admission residual equal to the original norm and giving $w_j = 1$ universally. As a consequence, the admitted directions form an orthonormal subset of the Fourier basis of $\mathbb{C}^q$, the GS process saturates at rank q, and the companion paper's numerical hypothesis becomes a theorem.

A secondary result concerns the conjugate pair $\{c, q-c\}$. The single-step Weil action satisfies $\rho_{q-c}(g)v_0 = \overline{\rho_c(g)v_0}$, and the k=3 fingerprint inherits a sign-reflection of the frequency index, $A_{q-c} = -A_c \pmod q$, for matched blocks. This shows that the two factor spans satisfy $V_n^{(q-c)} = \overline{V_n^{(c)}}$, recovering the structural identity of [2] from the Fourier structure and deriving the closure hypothesis of the companion paper.

2 The k=3 Weil fingerprint formula

Let q be an odd prime and let $c \in (\mathbb{Z}/q\mathbb{Z})^\times$. The Weil representation ρ_c of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ on $\mathbb{C}^q$ acts on the uniform vector $v_0 = (q^{-1/2}, \ldots, q^{-1/2})^\top$ by

$$(\rho_c(a,b,\gamma)\, v_0)_t = \frac{1}{\sqrt{q}} \exp\!\left(\frac{2\pi i c}{q}\big(\gamma + b(t-a)\big)\right), \qquad t \in \mathbb{Z}/q\mathbb{Z}. \tag{1}$$

This is the formula implemented in the spectral pipeline (function `weil_batch_lut`, line 180 of `spectral_012.py`): $\mathtt{arg}[i,t] = (\gamma_i + b_i(t - a_i)) \bmod q$, output $= \psi[\mathrm{arg}]/\sqrt{q}$ with $\psi[r] = \exp(2\pi i c r/q)$.

A generic block is parametrised by $(c_1, c_2, c_3) \in (\mathbb{Z}/q\mathbb{Z})^3$. For a shell element $g \in \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and a triple of generators (s_1, s_2, s_3), the k=3 path fingerprint is the elementwise product

$$\mathbf{f}(g, s_1, s_2, s_3) = \rho_{c_1}(g_1)v_0 \odot \rho_{c_2}(g_2)v_0 \odot \rho_{c_3}(g_3)v_0, \tag{2}$$

where $g_1 = gs_1$, $g_2 = g_1 s_2$, $g_3 = g_2 s_3$ and $\odot$ is the elementwise product. Writing $g_j = (a_j, b_j, \gamma_j)$ for the Heisenberg coordinates of g_j, the t-th component of $\mathbf{f}$ is

$$\mathbf{f}_t = \frac{1}{q^{3/2}} \exp\left(\frac{2\pi i}{q} \sum_{j=1}^{3} c_j\big(\gamma_j + b_j(t - a_j)\big) \right). \tag{3}$$

Separating the t-dependent and t-independent parts gives the key factorisation.

Lemma 1 (Fingerprint as Fourier mode)**.** *Let* $A = \sum_{j=1}^{3} c_j b_j \bmod q$ *and* $B = \sum_{j=1}^{3} c_j(\gamma_j - a_j b_j) \bmod q$. *Then*

$$\mathbf{f}_t = \frac{e^{2\pi i B/q}}{q^{3/2}}\, e^{2\pi i A t/q} = \frac{e^{2\pi i B/q}}{q} \cdot \frac{e^{2\pi i A t/q}}{\sqrt{q}}.$$

Equivalently, writing $\hat{u}_A = (e^{2\pi i A t/q}/\sqrt{q})_{t=0}^{q-1}$ *for the unit-norm discrete Fourier mode at frequency* A,

$$\mathbf{f} = \frac{e^{2\pi i B/q}}{q}\, \hat{u}_A.$$

Proof. Expand the exponent of (3): $\sum_j c_j(\gamma_j + b_j(t - a_j)) = \big(\sum_j c_j b_j\big)t + \sum_j c_j(\gamma_j - a_j b_j) = At + B$. Hence $\mathbf{f}_t = q^{-3/2} \exp(2\pi i(At + B)/q) = (e^{2\pi i B/q}/q) \cdot (e^{2\pi i A t/q}/\sqrt{q}) = (e^{2\pi i B/q}/q)\,(\hat{u}_A)_t$. □

Note that $\|\hat{u}_A\| = 1$ (unit norm) and $\|\mathbf{f}\| = 1/q$. The frequency index $A \in \mathbb{Z}/q\mathbb{Z}$ encodes all t-dependence of $\mathbf{f}$; the phase B and the prefactor $1/q$ are scalar multiples that do not affect linear independence.

3 Orthogonality and the admission residual

Lemma 2 (Fourier orthogonality)**.** *For* $A, A' \in \mathbb{Z}/q\mathbb{Z}$,

$$\langle \hat{u}_A, \hat{u}_{A'} \rangle = \frac{1}{q} \sum_{t=0}^{q-1} e^{2\pi i (A - A')t/q} = \delta_{A,A'}.$$

Proof. For $A = A'$: each term is 1, sum equals q, divided by q gives 1. For $A \neq A'$: the sum is a complete geometric progression over q terms with ratio $e^{2\pi i(A-A')/q} \neq 1$, which is zero. □

Corollary 3 (Fingerprint inner products)**.** *Let* $\mathbf{f}$ *and* $\mathbf{f}'$ *be fingerprints at frequencies* A *and* A' *with global phases* B *and* B'*. Then*

$$\langle \mathbf{f}, \mathbf{f}' \rangle = \frac{e^{2\pi i(B'-B)/q}}{q^2}\, \delta_{A,A'}.$$

In particular: (i) if $A \neq A'$*,* $\langle \mathbf{f}, \mathbf{f}' \rangle = 0$ *(orthogonal); (ii) if* $A = A'$*,* $\mathbf{f}' = e^{2\pi i(B'-B)/q}\mathbf{f}$ *(scalar multiple, linearly dependent).*

Proof. Write $\mathbf{f} = (e^{2\pi iB/q}/q)\,\hat{u}_A$ and $\mathbf{f}' = (e^{2\pi iB'/q}/q)\,\hat{u}_{A'}$. By Lemma 2: $\langle \mathbf{f}, \mathbf{f}' \rangle = e^{2\pi i(B'-B)/q} q^{-2} \langle \hat{u}_A, \hat{u}_{A'} \rangle = e^{2\pi i(B'-B)/q} q^{-2} \delta_{A,A'}$. □

We can now state and prove the main result.

Theorem 4 (Universal admission orthogonality)**.** *Let* $\mathcal{V}_{n-1}$ *be the Gram–Schmidt span of the Weil fingerprint vectors admitted at BFS depths* $0, \ldots, n-1$*. For any fingerprint* $\mathbf{f}$ *at depth* n *that is linearly independent of* $\mathcal{V}_{n-1}$ *(i.e.* $\mathbf{f} \notin \mathcal{V}_{n-1}$*), the normalised Gram–Schmidt admission residual satisfies*

$$w_j := \frac{\|w_j^{\mathrm{GS}}\|}{\|\mathbf{f}\|} = 1,$$

where $w_j^{\mathrm{GS}} = \mathbf{f} - P_{\mathcal{V}_{n-1}}\mathbf{f}$ *is the residual after projection onto* $\mathcal{V}_{n-1}$*.*

Proof. By Lemma 1, $\mathbf{f} = (e^{2\pi iB/q}/q)\,\hat{u}_A$ for some frequency A. The span $\mathcal{V}_{n-1}$ is the linear span of the previously admitted basis directions, each of which is a unit-norm Fourier mode $\hat{u}_{A'}$ for some frequency $A' \in S_{n-1}$ (the set of frequencies seen before depth n). Because $\mathbf{f} \notin \mathcal{V}_{n-1}$, the frequency A is not in S_{n-1}. By Corollary 3(i), $\langle \mathbf{f}, \hat{u}_{A'} \rangle = 0$ for all $A' \in S_{n-1}$. Therefore $P_{\mathcal{V}_{n-1}}\mathbf{f} = 0$ and $w_j^{\mathrm{GS}} = \mathbf{f}$. Hence $\|w_j^{\mathrm{GS}}\| = \|\mathbf{f}\|$, so $w_j = 1$. □

Corollary 5 (Orthonormal admitted basis)**.** *The Gram–Schmidt process on the Weil fingerprint orbit of* $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *produces, upon admission, a set of orthonormal vectors* $\{\hat{u}_{A_1}, \hat{u}_{A_2}, \ldots\}$ *indexed by the distinct frequencies* $A_j \in \mathbb{Z}/q\mathbb{Z}$ *encountered in BFS order. The process saturates at rank* q *once all* q *frequencies have been seen.*

Proof. Each admitted direction is the normalized residual $w_j^{\mathrm{GS}}/\|w_j^{\mathrm{GS}}\| = \mathbf{f}/\|\mathbf{f}\| = \hat{u}_A$. Two admitted directions $\hat{u}_A$ and $\hat{u}_{A'}$ with $A \neq A'$ are orthogonal by Lemma 2. Since $|\mathbb{Z}/q\mathbb{Z}| = q$, at most q distinct frequencies exist, so rank $\leq q$. Rank q is attained once all q Fourier modes $\hat{u}_0, \ldots, \hat{u}_{q-1}$ have been admitted, which occurs in finite BFS time because the Cayley graph is connected and the group acts transitively on frequencies through the b-components. □

Remark 1 (Numerical validation)**.** *Theorem 4 explains the numerical observation of the companion paper [?]: $w_j = 1.000\,000$ exactly for every admitted direction at $q \in \{29, 61, 101, 151\}$ and multiple generic blocks, with standard deviation zero. The agreement is exact (not merely approximate) because the orthogonality of Lemma 2 holds exactly over $\mathbb{Z}/q\mathbb{Z}$.*

4 The conjugate pair: matched blocks and sign-reflected frequencies

For a prime q and $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, the single-step Weil action satisfies

$$\rho_{q-c}(a, b, \gamma)v_0 = \overline{\rho_c(a, b, \gamma)v_0}, \tag{4}$$

since replacing c by $q - c \equiv -c$ conjugates all phases in (1).

For the k=3 fingerprint of a matched block (c_1, c_2, c_3) and its conjugate $(q - c_1, c_2, c_3)$ (same c_2, c_3 and same path elements g_1, g_2, g_3):

$$\mathbf{f}_t^{(q-c_1)} = \frac{e^{2\pi i B^{(q-c_1)}/q}}{q}\,(\hat{u}_{A^{(q-c_1)}})_t, \qquad A^{(q-c_1)} = -c_1 b_1 + c_2 b_2 + c_3 b_3. \tag{5}$$

Defining the frequency shift $\Delta = c_2 b_2 + c_3 b_3$, the two frequency indices are $A^{(c_1)} = c_1 b_1 + \Delta$ and $A^{(q-c_1)} = -c_1 b_1 + \Delta$. The mean is Δ and the difference is $\pm c_1 b_1$.

Proposition 6 (Conjugate frequency symmetry)**.** *For matched blocks (c_1, c_2, c_3) and $(q - c_1, c_2, c_3)$ sharing the same path elements, the two frequency indices satisfy*

$$A^{(q-c_1)} = -A^{(c_1)} + 2\Delta \pmod q, \qquad \Delta = c_2 b_2 + c_3 b_3.$$

When $c_2 = c_3 = 0$ (pure single-character blocks), $\Delta = 0$ and $A^{(q-c_1)} = -A^{(c_1)}$: the two blocks sample sign-reflected frequencies, so $V_n^{(q-c_1)} = \overline{V_n^{(c_1)}}$ (complex-conjugate subspaces).

Proof. Direct substitution into (5) with $c_2 = c_3 = 0$: $A^{(q-c_1)} = -c_1 b_1 = -(c_1 b_1) = -A^{(c_1)}$. The span of $\{\hat{u}_A : A \in S_{c_1}(n)\}$ is the conjugate of $\{\hat{u}_{-A} : A \in S_{c_1}(n)\}$ (since $\overline{\hat{u}_A} = \hat{u}_{-A}$), so $V_n^{(q-c_1)} = \overline{V_n^{(c_1)}}$. □

Remark 2 (General blocks and the O17 identity)**.** *For general blocks (c_1, c_2, c_3) with $c_2, c_3 \neq 0$, the frequency shift $\Delta \neq 0$ and the sign-reflection is not exact. However, the COUNTING identity $\sigma_{q-c}(n) = \sigma_c(n)$ established in* O17 *[2] (Theorem 2.1 and Corollary 2.3) shows that the two blocks admit the same number of independent directions at every shell, regardless of the block parameters. The present Fourier analysis shows why: every admitted direction is a Fourier mode $\hat{u}_A$, and the count depends only on how many distinct frequencies A appear in the BFS, which is invariant under the sign-reflection $c_1 \to q - c_1$ because multiplication by -1 is a bijection on $\mathbb{Z}/q\mathbb{Z}$. The* O17 *identity $\delta_{q-c} = \delta_c$ is therefore a corollary of Theorem 4.*

5 Flat Schmidt spectrum and $S_{\text{ent}} = \log r_{\text{pair}}$

We now derive the companion paper's main result analytically.

Theorem 7 (Flat Schmidt spectrum on the residual support)**.** *Let $\{c, q-c\}$ be a conjugate pair with matched blocks $(c, 0, 0)$ and $(q-c, 0, 0)$. Let $S_c(\infty) \subset \mathbb{Z}/q\mathbb{Z}$ be the full set of frequencies that appear in the BFS orbit (with $|S_c(\infty)| = q$ at saturation). At BFS depth n, write $R(n) := |S_c(n)|$ for the admitted rank, and define*

$$S_c^{\text{res}}(n) := S_c(\infty) \setminus S_c(n), \qquad r_{\text{pair}}(n) := |S_c^{\text{res}}(n)| = R_\infty - R(n),$$

$$V_{\text{res}}^{(c)}(n) := \text{span}\big\{\hat{u}_A : A \in S_c^{\text{res}}(n)\big\},$$
$$V_{\text{res}}^{(q-c)}(n) := \text{span}\big\{\hat{u}_{-A} : A \in S_c^{\text{res}}(n)\big\} = \overline{V_{\text{res}}^{(c)}(n)}.$$

*Here $r_{\text{pair}}(n)$ coincides with the residual rank of the companion paper [**?**]: $r_{\text{pair}}(n) = R_\infty - R(n)$, which is non-increasing. The residual bipartite fibre $K_c^{\text{res}}(n) = V_{\text{res}}^{(c)}(n) \otimes V_{\text{res}}^{(q-c)}(n)$ carries the maximally entangled state*

$$|\Omega_n^{\text{res}}\rangle = \frac{1}{\sqrt{r_{\text{pair}}(n)}} \sum_{A \in S_c^{\text{res}}(n)} |\hat{u}_A\rangle \otimes |\hat{u}_{-A}\rangle.$$

The Schmidt weights are all equal to $1/r_{\text{pair}}(n)$, and the reduced entropy is

$$S_{\text{ent}}(n) = \log r_{\text{pair}}(n).$$

Proof. The residual modes $\{\hat{u}_A : A \in S_c^{\mathrm{res}}(n)\}$ are orthonormal by Lemma 2 (distinct frequencies). Proposition 6 with $c_2 = c_3 = 0$ gives $S_{q-c}(n) = -S_c(n)$ (sign-reflection of the admitted sets), so the residual sets also satisfy $S_{q-c}^{\mathrm{res}}(n) = -S_c^{\mathrm{res}}(n)$ (since -1 is a bijection on $\mathbb{Z}/q\mathbb{Z}$ and $-S_c(\infty) = S_{q-c}(\infty) = S_c(\infty)$). Hence $V_{\mathrm{res}}^{(q-c)}(n) = \overline{V_{\mathrm{res}}^{(c)}(n)}$. The natural maximally entangled state on $V_{\mathrm{res}} \otimes \overline{V_{\mathrm{res}}}$ is $|\Omega_n^{\mathrm{res}}\rangle$ as stated. Its Schmidt decomposition has $r_{\mathrm{pair}}(n)$ terms with equal coefficients $1/\sqrt{r_{\mathrm{pair}}(n)}$, giving reduced state $r_{\mathrm{pair}}(n)^{-1}\mathbf{1}_{r_{\mathrm{pair}}(n)}$ and $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$. □

Remark 3 (General blocks and the scope of the result)**.** *For general independently sampled blocks (c, c_2, c_3) and $(q-c, c_2', c_3')$ with $(c_2, c_3) \neq (0,0)$ or $(c_2', c_3') \neq (c_2, c_3)$, the residual frequency sets $S_c^{\mathrm{res}}(n)$ and $S_{q-c}^{\mathrm{res}}(n)$ are not in general sign-reflections of each other, so $V_{\mathrm{res}}^{(q-c)}(n) \neq \overline{V_{\mathrm{res}}^{(c)}(n)}$ and the identification of Schmidt partners $\hat{u}_A \leftrightarrow \hat{u}_{-A}$ does not follow automatically. The* O17 *equal-count identity [2] guarantees $r_{\mathrm{pair}}(n)$ is the same for both factors (same residual dimension), but does not fix the pairing of modes. The flat Schmidt spectrum on the canonical pair observable [?] therefore remains conditional on a canonical block-parameter identification for the general case.*

6 Epistemic status update for the companion paper

Theorem 4 and Theorem 7 resolve the two items listed as open in the companion paper [?]:

1. *Analytic proof of $w_j = 1$ (previously numerical).* The identity $w_j = 1$ now follows from Theorem 4: the k=3 Weil fingerprint is a Fourier mode, and a new-frequency mode is orthogonal to the entire prior span. The numerical evidence at $q \in \{29, 61, 101, 151\}$ is an exact confirmation (not merely consistent with) this theorem.

2. *Purity of the conjugate pair (previously a hypothesis).* For matched blocks, Theorem 7 derives the joint maximally entangled state $|\Omega_n\rangle$ on $V \otimes \overline{V}$ from the Fourier mode structure and Proposition 6. The diagonal Schmidt pairing $\hat{u}_A \leftrightarrow \hat{u}_{-A}$ follows from the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ applied to the Fourier basis. Hypothesis 1 of the companion paper (closure of the conjugate pair) is therefore a theorem in the matched-block setting.

One item remains open: the extension of the matched-block purity argument to the independently sampled blocks of the canonical O25 pair ob-

servable [2]. There, c_2 and c_3 are drawn independently for the two factors, so the frequency sets $S_c(n)$ and $S_{q-c}(n)$ need not be exact sign-reflections. The *O17* counting identity ($\delta_{q-c} = \delta_c$) ensures equal rank at every shell, but the literal identification of Schmidt partners $\hat{u}_A \leftrightarrow \hat{u}_{-A}$ requires either a canonical block-parameter choice or an additional structural argument.

7 Conclusion

The admission residual $w_j = 1$ is not a coincidence of the pipeline or a numerical artefact: it is an exact algebraic identity. The k=3 Weil fingerprint of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is always a Fourier mode, and the Gram–Schmidt process on Fourier modes produces only orthogonal admissions. The fingerprint of a new BFS direction is already orthogonal to the entire span of prior admissions, because it carries a frequency not yet seen, and distinct Fourier frequencies are orthogonal.

For matched blocks, this implies that the conjugate pair $\{c, q-c\}$ is a closed pure system whose Schmidt decomposition is flat. The reduced entropy is $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ analytically, not merely by the observed vanishing of an admissibility deficit.

The key formula is Lemma 1: the k=3 fingerprint factors as a global phase times a single Fourier mode. This factorisation is a direct consequence of the linearity of the phase function in t, which in turn follows from the Schrödinger representation formula (1) for the Weil action on the uniform vector. The algebraic source of entanglement orthogonality is therefore the linear t-dependence of Weil phases, not a dynamical or spectral property.

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